Léo Navarro Chafloque

Curriculum vitæ

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Date of birth: 19 January 1999
Swiss nationality
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Mathematical interests

Algebraic geometry and commutative algebra in mixed characteristic, especially perfectoid and prismatic methods. Homotopy theory and its connections to these areas.

Education

Nov. 2022–present **École polytechnique fédérale de Lausanne (EPFL)**
PhD in mathematics. Advisor: Prof. Zsolt Patakfalvi.
Expected completion: summer 2027. Provisional thesis title: *Perfectoid and prismatic methods in mixed characteristic algebraic geometry.*

2020–2022 **Rheinische Friedrich-Wilhelms-Universität Bonn**
Master's degree in mathematics, awarded in 2022.
Focus on algebraic geometry and algebraic topology.

2019–2020 **École normale supérieure, Paris (ENS Ulm)**
Third year of the bachelor's programme, as an exchange student.

2017–2019 **École polytechnique fédérale de Lausanne (EPFL)**
First two years of the bachelor's programme in mathematics.
Bachelor's degree awarded by EPFL in 2020.

Mathematical works

Preprints

2026 **On perfectoid ideals**
Léo Navarro Chafloque.
Preprint, arXiv:2610.31964.

2026 **On Perfectoidization of Finite Algebras over a Perfectoid Ring**
Ryo Ishizuka and Léo Navarro Chafloque.
Preprint, arXiv:2606.12229.

2026 **Decomposition Theorem for Perfectoid Rings along General Ideals**
Ryo Ishizuka and Léo Navarro Chafloque.
Preprint, arXiv:2606.06241.

Talks

2 Oct. 2026 **Monodromy groups beyond constant slopes**
A Sprint on Local Systems, University of Wuppertal.

7 May 2026 **Group Law on Perfectoid Group Schemes**
Séminaire d'algèbre et de géométrie, Université de Neuchâtel.

30 Apr. 2026 **Perfectoidizations of finite algebras over a perfectoid ring**
BENDZ seminar in algebraic geometry, ETH Zürich.

Teaching experience

Spring 2023–2026	**Anneaux et corps (Rings and fields)** — EPFL Principal or co-principal teaching assistant; second-year BSc mathematics.
Autumn 2023, 2024, 2025	**Algebraic Geometry II: schemes and sheaves** — EPFL Principal teaching assistant.
Autumn 2022	**Structures algébriques** — EPFL Teaching assistant; first-year BSc mathematics.
Autumn 2018	**Physics: mechanics** — EPFL Teaching assistant; first-year BSc mathematics.

Teaching award

2023/2024	**Dean's Award for Excellence in Teaching** — EPFL Awarded for teaching assistance in Algebraic Geometry II.

Student supervision at EPFL

Spring 2025	**De Rham cohomology and singular cohomology** Milo Blum — master's project.
Spring 2024	**Cohomology computations** Léo Karlen — bachelor's project.
Spring 2023	**Incarnations of Galois groups** Héloïse Mansat — bachelor's project.

Seminar organisation

Organiser or co-organiser of the following graduate seminars at EPFL:

Autumn 2026	**Perfectoid and prismatic methods**
Autumn 2024	**p-adic cohomology**
Summer 2024	**Higher categories**
Spring 2024	**Integral models of homotopy types**
Autumn 2023	**Homotopy theory in algebraic geometry and vice-versa**
Spring 2023	**Higher Galois theory**
Spring 2023	**Relative elliptic curves**

Earlier work and activities

2022	**Master's thesis: About Deligne–Lusztig varieties of GL_3** Supervisor: Dr. Alexander Ivanov. Co-supervisor: Prof. Peter Scholze. Study of Deligne–Lusztig varieties of GL_3 over $\mathbb{F}_p$ and $\mathbb{Q}_p$.
Spring 2020	**Mémoire at ENS (bachelor's project)** *Actions of pure braid groups and string link groups* Supervisor: Prof. Geoffroy Horel.